

Effects of Director Angle Anchoring Conditions on the Dynamic Moduli of Heterogeneous Nematic Polymers

Eric P. Choate*, M. Gregory Forest†, Zhenlu Cui** and Lili Ju*

*Department of Mathematics, University of South Carolina, Columbia, SC 29210

†Mathematics, Biomedical Engineering, University of North Carolina, Chapel Hill, NC 27599-3250

**Mathematics and Computer Science, Fayetteville State University, Fayetteville, NC 28301

Abstract. We examine the linear viscoelastic response of heterogeneous nematic polymers to small amplitude oscillatory shear flow, paying special attention to the macroscopic influence of strong plate anchoring conditions. To predict the dynamic moduli, we model the system with Stokes hydrodynamic equations with viscous and nematic stresses coupled with orientational dynamics driven by the flow, an excluded volume potential, and an elasticity potential. First, we show that for special cases of normal and tangential anchoring, we recover explicitly solvable Leslie-Ericksen-Frank behavior. In this case we observe significant differences between the moduli for normal and tangential anchoring, including a two-to-three order of magnitude enhancement of the storage modulus for normal over tangential anchoring. Then, we turn to a numerical study of oblique anchoring conditions, which are more complicated due to the appearance of order parameter gradients at leading order. When the anchoring angle is near 45 degrees, we observe significantly different scaling behavior in the storage modulus for high frequencies compared to the behavior for normal or tangential anchoring. Furthermore, we find that for low frequencies, normal anchoring gives the largest storage modulus and tangential the smallest; however, we see a brief window of higher frequencies where the maximum storage modulus occurs at an oblique anchoring angle.

Keywords: Nematic liquid crystal polymers, Linear viscoelasticity, Strong anchoring

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INTRODUCTION

We seek to predict the storage and loss moduli of a nematic liquid crystal polymer solution by the application of a small amplitude oscillatory shear flow. The stress-strain relationship is derived after solving a coupled system for the fluid velocity $\mathbf{v}$ and the second moment tensor $\mathbf{M}$ of the molecular orientational probability density function f , or its traceless part, the orientation tensor $\mathbf{Q} = \mathbf{M} - \frac{1}{3}\mathbf{I}$. Before the flow is applied, if the dimensionless excluded volume parameter N is sufficiently high ($N > 3$), the system away from the boundary has a preferred nematic equilibrium $\mathbf{Q}_0 = s_0(\mathbf{n}_0\mathbf{n}_0 - \frac{1}{3}\mathbf{I})$, for the equilibrium order parameter $s_0 = \frac{1}{4}(1 + 3\sqrt{1 - \frac{8}{3N}})$ and an arbitrary major director $\mathbf{n}_0 = (\cos \psi_0, \sin \psi_0, 0)$. Although we allow 1D heterogeneity across the gap, if $\mathbf{Q}_0$ is the strong nematic anchoring conditions at both plates in the absence of flow, we see a uniform orientation $\mathbf{Q}(y) \equiv \mathbf{Q}_0$, effectively extending the anchoring condition across the entire gap. We now examine the effect of applying a small amplitude oscillatory shear flow to this system, and we observe the key role that ψ_0 plays in the prediction of the storage and loss moduli.

We use the tensor model and stress tensor from [3] and assume Stokes flow with no pressure gradients so that

$$\frac{\partial}{\partial t}\mathbf{M} = \mathbf{\Omega} \cdot \mathbf{M} - \mathbf{M} \cdot \mathbf{\Omega} + a(\mathbf{D} \cdot \mathbf{M} + \mathbf{M} \cdot \mathbf{D} - 2\mathbf{D} : \mathbf{M}_4) - 6\alpha[\mathbf{Q} - N(\mathbf{M} \cdot \mathbf{M} - \mathbf{M} : \mathbf{M}_4)] + \Delta\mathbf{M} \cdot \mathbf{M} + \mathbf{M} \cdot \Delta\mathbf{M} - 2\Delta\mathbf{M} : \mathbf{M}\mathbf{M}, \quad (1)$$

$$0 = \nabla \cdot \boldsymbol{\tau}, \quad \boldsymbol{\tau} = \mu_1(\mathbf{D} \cdot \mathbf{M} + \mathbf{M} \cdot \mathbf{D}) + \mu_2\mathbf{D} : \mathbf{M}_4 + \mu_3\mathbf{D} + \frac{3a}{s_0^2}\alpha[\mathbf{Q} - N(\mathbf{M} \cdot \mathbf{M} - \mathbf{M} : \mathbf{M}_4)] + \frac{1-a}{2s_0^2}\mathbf{M} \cdot \Delta\mathbf{M} - \frac{1+a}{2s_0^2}\Delta\mathbf{M} \cdot \mathbf{M} + \frac{a}{s_0^2}\Delta\mathbf{M} : \mathbf{M}_4 - \frac{1}{4s_0^2}(M_{kl,i}M_{kl,j} - \mathbf{M} : \nabla\mathbf{M}), \quad (2)$$

where we nondimensionalize with the gap width h , the Frank stress $\tau_F = \frac{v k T N \mathcal{L}^2 s_0^2}{8h^2}$, and the characteristic viscosity $\eta_0 = \frac{v k T s_0^2}{D_r}$ for the polymer number density v , the penetration depth $\mathcal{L}$ of the elasticity potential, the rotational diffusion rate D_r , $a = \frac{R^2-1}{R^2+1}$ for the molecular aspect ratio R , $\alpha = \frac{8h^2}{N\mathcal{L}^2}$, and $\mu_1 = \frac{3\zeta_1 D_r}{s_0^2}$, $\mu_2 = \frac{3\zeta_2 D_r}{s_0^2}$, and $\mu_3 = \frac{2\eta_s}{\eta_0} + \frac{3\zeta_3 D_r}{s_0^2}$ for the solvent viscosity η_s and the friction coefficients ζ_i in [3]. The characteristic timescale is $t_0 = \frac{\eta_0}{\tau_F}$ and the dimensionless rate-of-strain and vorticity tensors are $\mathbf{D} = \frac{1}{2}(\nabla\mathbf{v} + \nabla\mathbf{v}^T)$ and $\mathbf{\Omega} = \frac{1}{2}(\nabla\mathbf{v} - \nabla\mathbf{v}^T)$.

To highlight the role of the director angle ψ , we introduce a spectral representation of the orientation tensor: $\mathbf{Q} = s(\mathbf{n}_1\mathbf{n}_1 - \frac{1}{3}\mathbf{I}) + \beta(\mathbf{n}_2\mathbf{n}_2 - \frac{1}{3}\mathbf{I})$, where the order parameters s and β measure the birefringence relative to the directors $\mathbf{n}_1 = (\cos \psi, \sin \psi, 0)$, $\mathbf{n}_2 = (-\sin \psi, \cos \psi, 0)$, and $\mathbf{n}_3 = (0, 0, 1)$.

We drive the flow with the imposed stress boundary conditions $\tau_{xy}(y = \pm \frac{1}{2}) = Er \cos \omega t$ although imposing the plate velocity will yield the same moduli. The nondimensional *Ericksen number* Er is the ratio of the flow-induced viscous stress to the Frank stress, or $Er = \frac{\tau_0}{\tau_F}$ where τ_0 is the dimensional boundary stress. The appropriate asymptotic limit to examine the linear response is therefore the small Ericksen number limit, and so we expand s, β, ψ, v_x , and τ in terms of Er , and we find that for the Leslie tumbling parameter $\lambda_L = \frac{a(2+s_0)}{3s_0}$, the leading order system is

$$\frac{\partial s^{(1)}}{\partial t} = 2\alpha \left((6 - N(s_0 + 2))s^{(1)} - 2Ns_0(1 - s_0)\beta^{(1)} \right) + \frac{2}{N} \frac{\partial^2 s^{(1)}}{\partial y^2} - \frac{4(1-s_0)}{3} \frac{\partial^2 \beta^{(1)}}{\partial y^2} + \sin 2\psi_0 \frac{a(1+2s_0-3s_0^2)}{3} \frac{\partial v_x^{(1)}}{\partial y}, \quad (3)$$

$$\frac{\partial \beta^{(1)}}{\partial t} = -6\alpha N s_0 \beta^{(1)} + \frac{2(1-s_0)}{3} \frac{\partial^2 \beta^{(1)}}{\partial y^2} - \sin 2\psi_0 \frac{a(1-s_0)}{3} \frac{\partial v_x^{(1)}}{\partial y}, \quad (4)$$

$$\frac{\partial \psi^{(1)}}{\partial t} = -\frac{1}{2}(1 - \lambda_L \cos 2\psi_0) \frac{\partial v_x^{(1)}}{\partial y} + \frac{2+s_0}{3} \frac{\partial^2 \psi^{(1)}}{\partial y^2}, \quad (5)$$

$$0 = \frac{\partial \tau_{xy}^{(1)}}{\partial y}, \quad \tau_{xy}^{(1)} = \left(\frac{\mu_1(s_0+2)}{6} + \frac{\mu_2 s_0^2}{4} \sin^2 2\psi_0 + \frac{\mu_3}{2} \right) \frac{\partial v_x^{(1)}}{\partial y} + \frac{1}{2} (1 - \lambda_L \cos 2\psi_0) \frac{\partial^2 \psi^{(1)}}{\partial y^2} + \sin 2\psi_0 \frac{a}{2s_0^2} \left[\alpha((-6 + N(2 + s_0))s^{(1)} + (3 - N(1 + 2s_0))\beta^{(1)}) - \frac{\partial^2 s^{(1)}}{\partial y^2} + \frac{(1-s_0^2)}{3} \frac{\partial^2 \beta^{(1)}}{\partial y^2} \right]. \quad (6)$$

NORMAL AND TANGENTIAL ANCHORING

In [1], the authors examined analytically the special cases of normal ($\psi_0 = \frac{\pi}{2}$) and tangential anchoring ($\psi_0 = 0$). In these two cases, $\sin 2\psi_0 = 0$ so that the order parameter equations (3)-(4) decouple from the director angle and velocity equations (5)-(6). The order parameter system has the solutions $s^{(1)}(y, t) \equiv \beta^{(1)}(y, t) \equiv 0$, and so at leading order, the tensor model with normal or tangential anchoring reduces to dynamics similar to Leslie-Ericksen theory:

$$\frac{\partial \psi^{(1)}}{\partial t} = A \frac{\partial^2 \psi^{(1)}}{\partial y^2} + B \frac{\partial v_x^{(1)}}{\partial y}, \quad 0 = \frac{\partial \tau_{xy}^{(1)}}{\partial y}, \quad \tau_{xy}^{(1)} = -B \frac{\partial^2 \psi^{(1)}}{\partial y^2} + C \frac{\partial v_x^{(1)}}{\partial y}, \quad (7)$$

where $A = \frac{s_0+2}{3}$, $C = \frac{\mu_1(s_0+2)}{6} + \frac{\mu_3}{2}$, and $B = -\frac{1-\lambda_L}{2}$ for tangential anchoring, but $B = -\frac{1+\lambda_L}{2}$ for normal anchoring.

Since the system (7) is linear, and the plate driving conditions are sinusoidal in time, the standard method of determining the linear viscoelastic moduli is to suppress transients and seek a frequency-locked solution of the form $\psi^{(1)}(y, t) = \psi_1(y) \cos \omega t + \psi_2(y) \sin \omega t$, $v_x^{(1)}(y, t) = v_1(y) \cos \omega t + v_2(y) \sin \omega t$, and $\tau_{xy}^{(1)}(t) = \tau_1 \cos \omega t + \tau_2 \sin \omega t$. The general solution of this form introduces a new length scale $r^{-1} = \sqrt{\frac{2(AC+B^2)}{C\omega}}$, and the boundary conditions on ψ_1 and ψ_2 allow us to express the components of the upper plate velocity $V_i = v_i(\frac{1}{2})$ in terms of τ_1 and τ_2 :

$$V_1 = E_2 \tau_1 + E_1 \tau_2, \quad V_2 = -E_1 \tau_1 + E_2 \tau_2, \quad \text{where } E_1 = \frac{B^2 r}{C^2 \omega} \frac{\sinh r - \sin r}{\cosh r + \cos r}, \quad E_2 = \frac{1}{2C} - \frac{B^2 r}{C^2 \omega} \frac{\sinh r + \sin r}{\cosh r + \cos r}. \quad (8)$$

Thus, we have arrived at a relationship between the shear stress and the macroscopic strain $\gamma = \frac{2}{\omega}(V_1 \sin \omega t - V_2 \cos \omega t)$, which is all that is needed to identify the storage and loss moduli as, respectively,

$$G'(\omega) = \frac{\omega}{2} \frac{\tau_2 V_1 - \tau_1 V_2}{V_1^2 + V_2^2} = \frac{\omega}{2} \frac{E_1}{E_1^2 + E_2^2} = \frac{2B^2 C^2 \omega^2 r (\sinh r - \sin r)}{(8B^4 r^2 + C^2 \omega^2)(\cosh r + \cos r) - 16B^4 r^2 \cos r - 4B^2 C r \omega (\sin r + \sinh r)}, \quad (9)$$

$$G''(\omega) = \frac{\omega}{2} \frac{\tau_1 V_1 + \tau_2 V_2}{V_1^2 + V_2^2} = \frac{\omega}{2} \frac{E_2}{E_1^2 + E_2^2} = \frac{C^2 \omega^2 (C \omega (\cosh r + \cos r) - 2B^2 r (\sinh r + \sin r))}{(8B^4 r^2 + C^2 \omega^2)(\cosh r + \cos r) - 16B^4 r^2 \cos r - 4B^2 C r \omega (\sin r + \sinh r)}. \quad (10)$$

Asymptotically, the loss modulus $G''(\omega)$ exhibits nearly linear scaling behavior for both high frequencies $G''(\omega) = (C + \frac{B^2}{A})\omega$, as $\omega \rightarrow 0$, and low frequencies $G''(\omega) = C\omega$, as $\omega \rightarrow \infty$. The constant offset between the high and low frequency limits, $\frac{B^2}{A}$, is always positive but given the dependence of B on ψ_0 , it is much larger for normal anchoring than for tangential. We also find different asymptotic regimes for the storage modulus, but they have distinct scaling behaviors with respect to ω . For low frequencies, $G'(\omega) = \frac{B^2}{12A^2} \omega^2$, as $\omega \rightarrow 0$, and for high frequencies, $G'(\omega) = B^2 \sqrt{\frac{2C}{AC+B^2}} \sqrt{\omega}$, as $\omega \rightarrow \infty$. The factor $B^2 = \frac{(1+\lambda_L)^2}{4}$ means that G' for normal anchoring can be two to three orders of magnitude larger for tangential anchoring, as depicted in Figure 1.

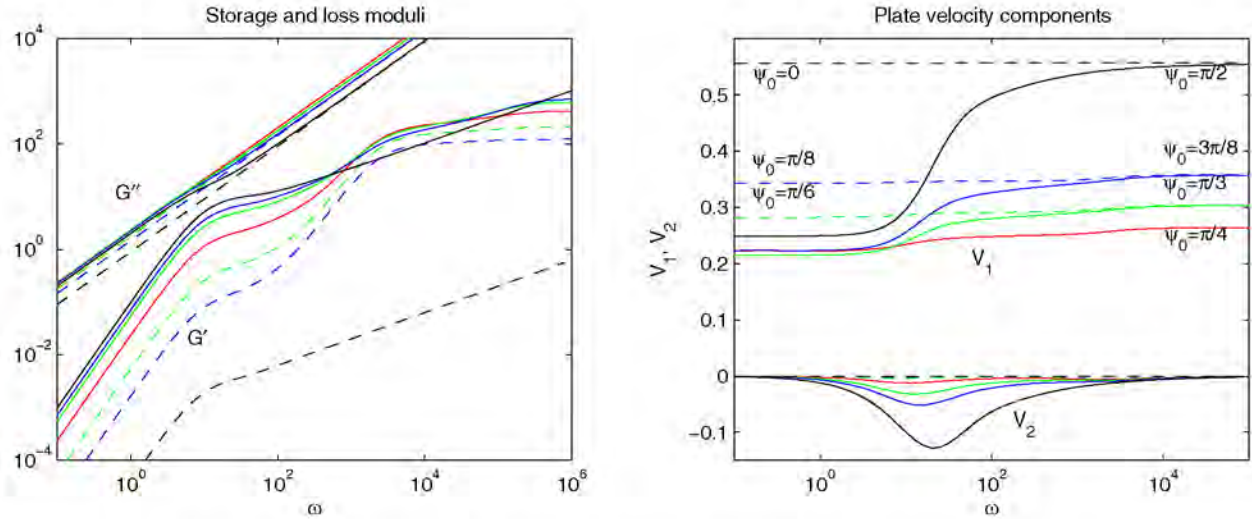


FIGURE 1. The storage (G') and loss (G'') moduli and the in-phase (V_1) and out-of-phase (V_2) components of the plate velocity for $\psi_0 = 0$ (black dashed), $\frac{\pi}{8}$ (blue dashed), $\frac{\pi}{6}$ (green dashed), $\frac{\pi}{4}$ (red), $\frac{\pi}{3}$ (green solid), $\frac{3\pi}{8}$ (blue solid), and $\frac{\pi}{2}$ (black solid).

OBLIQUE ANCHORING

Now we turn our attention to director angles that are neither tangential nor normal. A symmetry of the system (3)-(6) means that moduli for ψ_0 and $-\psi_0$ are equal. We solved the system numerically for a variety of anchoring angles over several decades of the frequency ω , and Figure 1 shows the resulting storage and loss moduli and the in-phase and out-of-phase components of the velocity of the upper plate in response to an imposed stress boundary condition.

The most striking qualitative effect of ψ_0 is in the storage modulus. For low frequencies, G' for oblique angles is $O(\omega^2)$, the same scaling behavior as normal and tangential anchoring, and G' increases as ψ_0 increases from 0 to $\frac{\pi}{2}$. However, for a brief window of frequencies, the largest value of G' occurs when ψ_0 is near $\frac{\pi}{4}$ (the red curve). Then, for higher frequencies, the storage modulus for oblique anchoring exhibits a different scaling than for normal or tangential anchoring with $G'(\omega) = O(1)$ as $\omega \rightarrow \infty$. This scaling is however consistent with the author's previous work [2] on the effect of the director angle degeneracy in monodomain systems although that work predicted that G' increased monotonically with ψ_0 for all frequencies.

For G'' , we see linear asymptotic regimes for both high and low frequencies for all anchoring angles. For low frequencies, the values of G'' are virtually indistinguishable for $\frac{\pi}{4} \leq \psi_0 \leq \frac{\pi}{2}$ with smaller values for ψ_0 near zero. However, for high frequencies the largest value of G'' occurs at $\psi_0 = \frac{\pi}{4}$ and the values for ψ_0 and $\frac{\pi}{2} - \psi_0$ nearly overlap with $\psi_0 = 0$ and $\frac{\pi}{2}$ providing the smallest value. This is evident in the in-phase portion of the plate velocity as well. For low frequencies, tangential anchoring has the fastest plate while the slowest plate has an anchoring angle near $\psi_0 = \frac{\pi}{3}$. However, as the frequency increases, the plates speed show very little change for angles less than $\frac{\pi}{4}$, but for larger angles, the plate speed increases towards the corresponding solution with the anchoring angle $\frac{\pi}{2} - \psi_0$.

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